

DevOps Exam Submission - ID Card Management System (Kubernetes)

Student Information

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- **Group:** B
- **Date:** June 17, 2026

Repository Information

- **GitHub URL:** <https://github.com/ChingHoir/Final-devop.git>
 - **Branch:** main
 - **Latest Commit:** [6145e98](#)
-

Task 1: Kubernetes Configuration (CHOICE_B - Kubernetes Pod)

YAML File Created

File: [id-card-management/kubernetes/minikube-deployment.yaml](#)

Pod Configuration Summary

Item	Value
Pod Name	e20221469-pod
Web Container	web-server (JDK 21 + NGINX + Spring Boot + PHP + SSH)
DB Container	mysql (MySQL 8.0)
Database Name	B-Huy_Chanchhinghoir-db
DB User/Password	root / Hello@123
NGINX Port	8080 (exposed as 8443 via Service)
SSH Port	22 (exposed as 2222 via Service)
Website NodePort	30843
SSH NodePort	30222

Verification Commands & Output

```
6145e98..9f9bec8 main -> main
PS D:\Year4\semester2\dev\Final> kubectl get pods
NAME                READY   STATUS    RESTARTS   AGE
e20221469-pod       2/2     Running   0           6m49s
PS D:\Year4\semester2\dev\Final>
```

Task 2: Web Container - PHP Modules Output

Command Executed

```
kubectl exec e20221469-pod -c web-server -- php -m
```

Output (saved to `php-modules.txt`)

```
PS D:\Year4\semester2\dev\Final> kubectl get pods
PS D:\Year4\semester2\dev\Final> kubectl exec e20221469-pod -c web-server -- php -m
[PHP Modules]
calendar
Core
ctype
date
exif
FFI
fileinfo
filter
ftp
gettext
hash
iconv
json
lexbor
libxml
openssl
pcntl
pcre
PDO
Phar
posix
random
readline
Reflection
session
shmop
sockets
sodium
SPL
standard
sysvmsg
sysvsem
sysvshm
tokenizer
uri
Zend OPcache
zlib

[Zend Modules]
Zend OPcache
```

File location: `php-modules.txt` (repository root)

Task 3: MySQL Container - Show Tables Output

Command Executed

```
kubectl exec e20221469-pod -c mysql -- mysql -u root -pHello@123 B-Huy_Chanchhinghoir-db -e "show tables;"
```

Output (saved to `mysql-tables.txt`)

```
PS D:\Year4\semester2\dev\Final> kubectl exec e20221469-pod -c mysql -- mysql -u root -pHello@123 B-Huy_Chanchhinghoir-db -e "show tables;"
mysql: [Warning] Using a password on the command line interface can be insecure.
Tables_in_B-Huy_Chanchhinghoir-db
profiles
templates
```

File location: `mysql-tables.txt` (repository root)

Task 4: Git Commands

```
# Add Kubernetes YAML file
git add id-card-management/kubernetes/minikube-deployment.yaml

# Add PHP modules output
git add php-modules.txt

# Add MySQL tables output
git add mysql-tables.txt

# Add Dockerfile and nginx config
git add id-card-management/Dockerfile id-card-management/nginx/nginx.conf

# Commit all files
git commit -m "add Kubernetes deployment, PHP modules, MySQL tables, and
configuration files"

# Push to GitHub
git push origin main
```

Commit Details

- **Commit Hash:** `6145e98`
 - **Commit Message:** "add Kubernetes deployment, PHP modules, MySQL tables, and configuration files"
-

Dockerfile Configuration

File: `id-card-management/Dockerfile`

The Dockerfile uses a multi-stage build:

1. **Build stage:** Maven 3.9 + JDK 21 to compile the Spring Boot application
2. **Runtime stage:** JDK 21 JRE with:
 - NGINX (web server, port 8080, proxies to Spring Boot on 8081)
 - PHP (for `php -m` command)
 - OpenSSH Server (port 22, root user with password `Hello@123`)
 - MySQL client (for waiting until MySQL is ready)

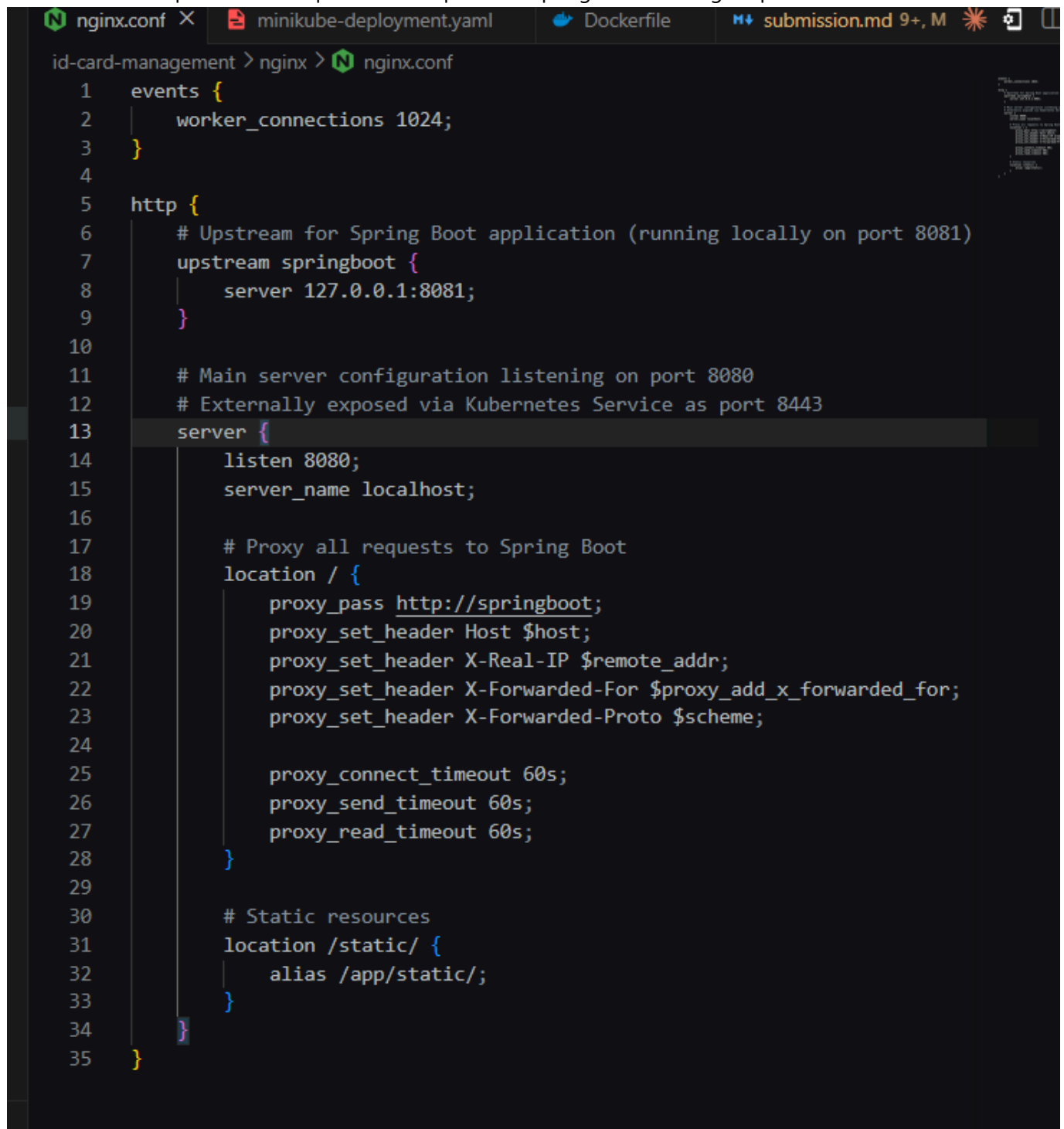
Docker Build Command

```
service "e20221403-service" deleted from default namespace
PS D:\Year4\semester2\dev\Final> cd id-card-management; minikube -p minikube docker-env --shell powershell | Invoke-Expression; docker build -t id-card-app:late
[+] Building 39.8s (18/18) FINISHED                                docker:default
=> [internal] load build definition from Dockerfile                0.0s
=> => transferring dockerfile: 1.41kB                             0.0s
=> [internal] load metadata for docker.io/library/maven:3.9-eclipse-temurin-21 1.0s
=> [internal] load metadata for docker.io/library/eclipse-temurin:21-jre 1.0s
=> [internal] load .dockerignore                                  0.0s
=> => transferring context: 2B                                     0.0s
=> [build 1/6] FROM docker.io/library/maven:3.9-eclipse-temurin-21@sha256:d7e7f57407437c014 0.0s
=> [internal] load build context                                0.0s
=> => transferring context: 2.71kB                                0.0s
=> [stage-1 1/6] FROM docker.io/library/eclipse-temurin:21-jre@sha256:194b590649efee5555de0 0.0s
=> CACHED [build 2/6] WORKDIR /app                               0.0s
=> CACHED [build 3/6] COPY pom.xml .                             0.0s
=> CACHED [build 4/6] RUN mvn dependency:go-offline -B           0.0s
=> CACHED [build 5/6] COPY src ./src                             0.0s
=> CACHED [build 6/6] RUN mvn clean package -DskipTests         0.0s
=> CACHED [stage-1 2/6] WORKDIR /app                             0.0s
=> CACHED [stage-1 3/6] COPY --from=build /app/target/id-card-management-1.0.0.jar app.jar 0.0s
=> [stage-1 4/6] RUN apt-get update && apt-get install -y nginx php openssl-se 37.3s
=> [stage-1 5/6] COPY nginx/nginx.conf /etc/nginx/nginx.conf   0.0s
=> [stage-1 6/6] RUN mkdir -p /var/run/sshd && echo "root:Hello@123" | chpasswd && 0.2s
```

NGINX Configuration

File: id-card-management/nginx/nginx.conf

NGINX listens on port 8080 and proxies all requests to Spring Boot running on port 8081:



```
id-card-management > nginx > nginx.conf
1  events {
2      worker_connections 1024;
3  }
4
5  http {
6      # Upstream for Spring Boot application (running locally on port 8081)
7      upstream springboot {
8          server 127.0.0.1:8081;
9      }
10
11     # Main server configuration listening on port 8080
12     # Externally exposed via Kubernetes Service as port 8443
13     server {
14         listen 8080;
15         server_name localhost;
16
17         # Proxy all requests to Spring Boot
18         location / {
19             proxy_pass http://springboot;
20             proxy_set_header Host $host;
21             proxy_set_header X-Real-IP $remote_addr;
22             proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
23             proxy_set_header X-Forwarded-Proto $scheme;
24
25             proxy_connect_timeout 60s;
26             proxy_send_timeout 60s;
27             proxy_read_timeout 60s;
28         }
29
30         # Static resources
31         location /static/ {
32             alias /app/static/;
33         }
34     }
35 }
```

Submission Checklist

- ☒ Kubernetes Pod YAML file created (`minikube-deployment.yaml`)
- ☒ Kubernetes Service YAML file created (same file)
- ☒ YAML file committed to GitHub
- ☒ PHP modules saved to `php-modules.txt` at repo root
- ☒ `php-modules.txt` committed to GitHub
- ☒ MySQL tables saved to `mysql-tables.txt` at repo root
- ☒ `mysql-tables.txt` committed to GitHub
- ☒ Docker image built successfully

- ☒ Pod is running with 2/2 containers
- ☒ All files pushed to GitHub repository
- ☒ Repository URL: <https://github.com/ChingHoir/Final-devop.git>

Application Access

Service	URL/Command
Website	http://localhost:30843 (via NodePort)
SSH Access	<code>ssh root@localhost -p 30222</code> (password: Hello@123)
Database	<code>B-Huy_Chanchhinghoir-db</code> on 127.0.0.1:3306

Date: June 17, 2026
Submitted by: HUY Chanchhinghoir (ID: 20221469 - Group B)